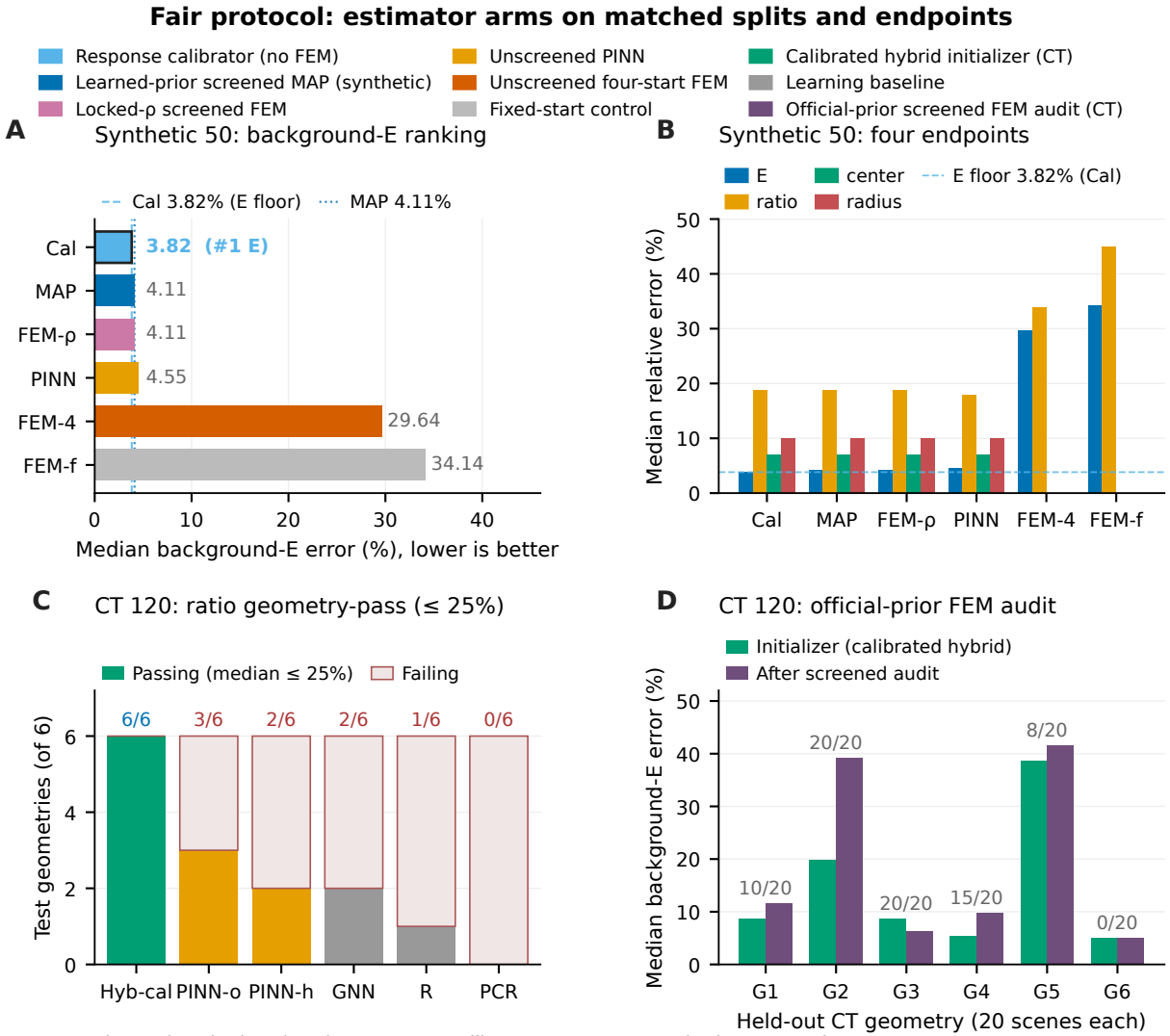

Key, panels A–B (synthetic 50): Cal = Response calibrator · MAP = Learned-prior screened MAP · FEM-ρ = Locked-ρ screened FEM · PINN = Unscreened PINN inverse · FEM-4 = Four-start FEM (no screen) · FEM-f = Fixed-start FEM

Panel B: center/radius share the predicted sphere and are not defined for FEM-4 / FEM-f.

Key, panel C (CT 120): Hyb-cal = Calibrated hybrid initializer (17.77%) · PINN-o = PINN, official start · PINN-h = PINN, hybrid-initializer start · GNN = Raw MeshGNN (26.95%) · R = Ridge (same protocol) · PCR = PCA–Ridge (same protocol)

Key, panel D (CT 120 official-prior audit): G1 = 58db559b, G2 = 62010c56, G3 = a417d13e, G4 = ab578f1b, G5 = db38a9c3, G6 = fac5cdc8

Panel D: n/20 above each pair = accepted refinements; pooled median 8.55% → 11.70% (73/120 accepted); ratio fixed at the locked 17.77%.